# The Polyakov–Wiegmann Identity

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Let $\Sigma$ be a Riemann surface, and choose a 3-dimensional manifold $B$ such that $\partial B = \Sigma$. For a Lie-group-valued function $f$ on $\Sigma$, extended to $B$, the WZW action is defined as

$$S_\Sigma(f) = -\frac{ik}{4\pi}\int_\Sigma \mathrm{Tr}(f^{-1}\partial f \wedge f^{-1}\bar\partial f) - \frac{ik}{12\pi}\int_B \mathrm{Tr}(f^{-1}df \wedge f^{-1}df \wedge f^{-1}df).$$

The Polyakov–Wiegmann identity asserts that

$$S_\Sigma(fg) = S_\Sigma(f) + S_\Sigma(g) - \frac{ik}{2\pi}\int_\Sigma \mathrm{Tr}(f^{-1}\partial f \wedge \bar\partial g\, g^{-1}).$$

Denote $S_0(f) = \int_\Sigma \mathrm{Tr}(f^{-1}\partial f \wedge f^{-1}\bar\partial f)$ and $\Gamma(f) = \int_B \mathrm{Tr}(f^{-1}df \wedge f^{-1}df \wedge f^{-1}df)$.

By the Leibniz rule we have $(fg)^{-1}\partial(fg) = g^{-1}f^{-1}\partial f\, g + g^{-1}\partial g$, and using the cyclicity of the trace:

$$\begin{aligned} S_0(fg) &= S_0(f) + S_0(g) + \int_\Sigma \mathrm{Tr}(f^{-1}\partial f \wedge \bar\partial g\, g^{-1} + \partial g\, g^{-1} \wedge f^{-1}\bar\partial f),\\ \Gamma(fg) &= \Gamma(f) + \Gamma(g) + 3\int_B \mathrm{Tr}(f^{-1}df \wedge f^{-1}df \wedge dg\, g^{-1} + f^{-1}df \wedge dg\, g^{-1} \wedge dg\, g^{-1}). \end{aligned}$$

So we need to show that

$$\int_B \mathrm{Tr}(f^{-1}df \wedge f^{-1}df \wedge dg\, g^{-1} + f^{-1}df \wedge dg\, g^{-1} \wedge dg\, g^{-1}) = \int_\Sigma \mathrm{Tr}(-f^{-1}\partial f \wedge \bar\partial g\, g^{-1} + \partial g\, g^{-1} \wedge f^{-1}\bar\partial f).$$

The right-hand side equals $-\int_\Sigma \mathrm{Tr}(f^{-1}df \wedge dg\, g^{-1})$, since terms such as $\partial f \wedge \partial g$ are multiples of $dz \wedge dz$ and hence vanish.

Finally,

$$d\big(\mathrm{Tr}(f^{-1}df \wedge dg\, g^{-1})\big) = \mathrm{Tr}(-f^{-1}df \wedge f^{-1}df \wedge dg\, g^{-1} - f^{-1}df \wedge dg\, g^{-1} \wedge dg\, g^{-1}).$$

By Stokes' formula we complete the proof.